Data Science Project Topics Presentation

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Outlook

Topic 1:
How does **Trump's news** coverage move the **stock markets**?

Topic 2:
Effects of **Weather** on different modes of **Transport** in Kiel

Topic 3:
Effects of **account age** and **played hours** on the value of **counter-strike inventories**

Analyses of the correlation between Trump news coverage and market movements among different volatility classes

yfinance
Pythonic access to market data

News source: The Guardian (API),
categories: trade policy, geopolitics, domestic politics

Market data: yfinance (API)
MSCI World, Gold, Bitcoin (low to extreme volatility)

Time period: Trump's current legislative period

Research questions

1. To what extent does the **frequency of Trump-related news coverage**, categorized by topic (trade policy, geopolitics, domestic politics), correlate with **short-term price changes** (1–3 trading days) in MSCI World, Gold, and Bitcoin?
2. How do MSCI World, Gold, and Bitcoin differ in the **direction and magnitude of their price responses** to identical Trump-related news events?
3. How does the sentiment **polarity** (positive vs. negative) of Trump-related Guardian articles within each news category relate to the **direction and magnitude** of subsequent price movements across all three asset classes?
4. To what degree do **simultaneous spikes across multiple news categories** amplify market reactions compared to isolated single-category spikes, and which asset class shows the greatest sensitivity to this **multi-topic overlap effect**?
5. Is there a news threshold, like a minimum number of news articles on one topic before the markets show a response on news? If yes what is that threshold?
6. How does the **sentiment** (positive, negative, neutral) of Trump-related news within each category influence the **direction and magnitude** of short-term price reactions in MSCI World, Gold, and Bitcoin?
7. Which of the three news categories (trade policy, geopolitics, domestic politics) generates the **highest average daily article volume** in The Guardian during Trump's current legislative period, and does the most frequently covered category also show the **strongest association with price movements**?

Analysing the effects different Weather Conditions have on the usage of Car, Bus and Bike in Kiel.

Weather Data: Open Meteo (API)
High Resolution weather data

Bike Data: Nextbike(API), mobilithek.info (API)
Data on rentals of Sprottenflotte Bikes, Data from official bike counting stations in Kiel

Bus Data: opendata.schleswig-holstein (dataset)
Data on Bus routes and Usage by the KVG

Car Data: bast.de (API)
Data from official car counting stations in Kiel

Research questions

How do **weather** conditions influence the **choice of transport** in Kiel, and does "bad weather" cause a measurable shift from cycling to cars or public transit?

Does **rain intensity** (light <2mm, moderate 2-10mm, heavy >10mm) have a **threshold effect on cycling** rates, or is the reduction gradual?

How does **temperature** correlate with modal split across seasons — is there a specific temperature threshold (e.g. below 5°C) where **cycling drops sharply** and KVG ridership spikes?

Do **weekday** commuters and **weekend** recreational cyclists **react differently to bad weather**, suggesting that cycling necessity vs. choice plays a role in modal shift?

Which weather variable — temperature, precipitation, wind speed, or snowfall — is the **strongest single predictor** of reduced bike usage at Kiel's counting stations?

Research questions

Is there a "compensation effect" where a **drop in Nextbike** (shared bikes) during bad weather is **larger than** the **drop at fixed counting stations**, suggesting casual riders avoid bikes more than regular cyclists?

How do **extreme weather events** (storms, ice, heavy snow) affect the three transport modes differently — do they **suppress all modes equally** or does one mode prove more resilient?

Does the weather-modal split relationship **change over the years 2020–2024**, potentially reflecting growing cycling infrastructure or post-COVID behavior shifts in Kiel?

Are there differences in how weather impacts transport choices across **different months or seasons**, and do these differences suggest that certain weather conditions are more tolerable for cycling during specific times of the year?

Analysing the relationship between Counter-Strike Item-Prices and playerprofiles and inventory-value with regards to account age and played hours

Steam Account data: steampowered (API)
High Volume of Data on individual Profiles

Market Data: steamcommunity(API)
Data on market prices of different Items

Research questions

How does Steam **account age** influence the a**verage inventory value** of Counter-Strike players, and is the relationship linear or non-linear?

What is the relationship between **total playtime** and **inventory value** in Counter-Strike, and does additional playtime contribute differently to inventory growth at higher experience levels?

To what extent does **inventory value** correlate with **in-game rank**, and can inventory value serve as a predictor of competitive performance?

How does the **frequency of gameplay sessions** (activity density) correlate with **inventory value** compared to total lifetime hours?

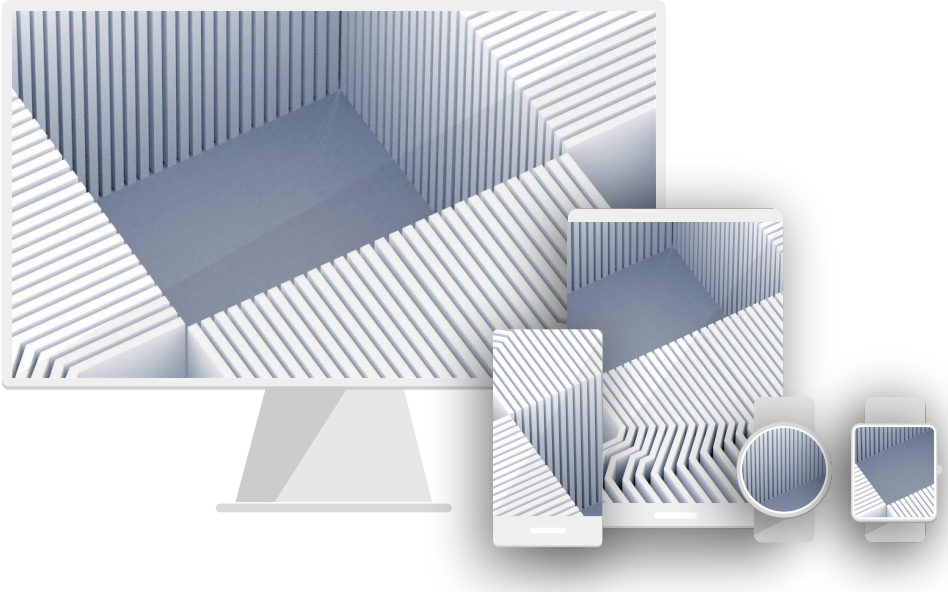
To what extent does **"Steam Profile Level"** and **community interaction** (e.g., number of badges, comments, or friends) predict the rarity and **value** of a Counter-Strike **inventory**?

Research questions

How does the **ratio** of **"Consumable"** items (Cases, Stickers, Graffitis) to **"Skin"** items (Weapons, Knives) change as an **account ages**?

How does **VAC ban** history or account standing correlate with **inventory composition** — specifically, do flagged or recently banned accounts show atypical item distributions compared to clean accounts?

(Advanced) Can a **machine learning model** trained on account age, playtime, Steam level, friend count, and activity density accurately **predict inventory value** bracket, and which features carry the most predictive weight?

Thank you for your Attention

Questions?